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Diagnostics-VF.CONCEPT

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1 Introduction

This document is a function specification for the control of the Diagnostic system for V300ZEFIRO for customer Trenitalia (VZES Project).

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:

- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

CD	Circuit Diagram
CCU	Central Computing Unit
CCU-D	Central Computing Unit - Diagnostic
CCU-O	Central Computing Unit - Operational
CSS	Common Software Structures (Operation system of CCU-O)
ECN	Ethernet Consist Network
EVENT	Diagnostics that will be stored in the Event database within TCMS system.
GUI	Graphical User Interface
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Interface Display Unit
IPT	Internet Protocol - Train
MAVIS	Maintenance of Vehicles using Information and Statistics
MTVD	MITRAC CC Tools Version control and Download
ODBS	Onboard Diagnostic Database System
PCL	Process Client
TCL	Tool Client
TCMS	Train Control and Management System
TDS	Train Diagnostic System
TSSP	Train Subscription Service Provider
VRS	Version Reporting System
WBS	Work Breakdown Structure

2 System Overview

The purpose of the TCMS TDS (Train Diagnostic System) is to ascertain and provide the current status of the vehicle by gathering status data among the vehicle systems. Diagnostic messages are generated from both TCMS equipment as well as sub-suppliers equipment and stored by the diagnostic system. In case an event occurs which causes an operational impact to the vehicle it will be presented to the user combined with user advice.

2.1 System Configuration

The on-board part of the train diagnostic system, TDS, is a sub-system of the TCMS. The TCMS TDS located in CCU-D collects, stores and acts upon relevant information about events, faults and vehicle status from the connected systems. The event information stored in the TDS database can be presented by the TCMS IDU and can be uploaded to an off-board maintenance and service system for long term storage and additional off line analysis.

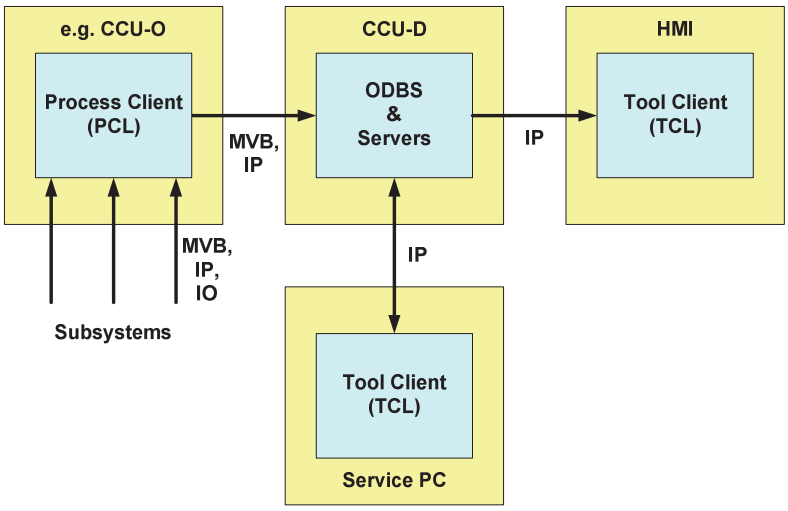


Figure: Train Diagnostic System (TDS)

For failure mode and redundancy concept refer to SFRS TCMS Operational.

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3 Functional requirements

3.1 Diagnostics

3.1.1 Function General requirements

3.1.1.1 FunctionDesign

The TCMS diagnostic system stores the events defined in other SFRS documents. As events are detected both in the TCMS, as well as in the propulsion system, the scope split is defined.

3.1.1.2 Requirements

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3.3.4.2 Requirements

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4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Inputs:

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5 References

3EVT12158-VZI TDS Process

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6 Revision Log

6.1 Revision log 3ECP413922-1001__

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Editor:

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